

Python Assignment – 27

1: Write a Python program to implement a class named BookStore with the following specifications:

- The class should contain two instance variables:
 - Name (Book Name)
 - Author (Book Author)
- The class should contain one class variable:
 - NoOfBooks (initialize it to 0)
- Define a constructor (`__init__`) that accepts Name and Author and initializes instance variables.
- Inside the constructor, increment the class variable NoOfBooks by 1 whenever a new object is created.
- Implement an instance method:
 - Display() – should display book details in the format:
<BookName> by <Author>. No of books: <NoOfBooks>

Example usage:

```
Obj1 = BookStore("Linux System Programming", "Robert Love")
```

```
Obj1.Display()
```

```
# Linux System Programming by Robert Love. No of books: 1
```

```
Obj2 = BookStore("C Programming", "Dennis Ritchie")
```

```
Obj2.Display()
```

```
# C Programming by Dennis Ritchie. No of books: 2
```

Answer :

```
class BookStore:
    NoOfBooks = 0
    def __init__(self, Book_Name, Book_Author):
        self.Book_Name = Book_Name           #Instance Varibale
        self.Book_Author = Book_Author       #Instance Varibale

        NoOfBooks = NoOfBooks+1

    def Display(self):
        print(f"{self.Book_Name} by {self.Book_Author}. No of books: {BookStore.NoOfBooks}")

def main():
    print("--BOOK STORE DETAILS--")

    Obj1 = BookStore("Linux System Programming", "Robert Love")
    Obj1.Display()

    Obj2 = BookStore("C Programming", "Dennis Ritchie")
    Obj2.Display()
```

```
if __name__ == "__main__":
    main()
```

2: Write a Python program to implement a class named BankAccount with the following requirements:

- The class should contain two instance variables:
 - Name (Account holder name)
 - Amount (Account balance)
- The class should contain one class variable:
 - ROI (Rate of Interest), initialized to 10.5
- Define a constructor (`__init__`) that accepts Name and initial Amount.
- Implement the following instance methods:
 - `Display()` – displays account holder name and current balance
 - `Deposit()` – accepts an amount from the user and adds it to balance
 - `Withdraw()` – accepts an amount from the user and subtracts it from balance
(Ensure withdrawal is allowed only if sufficient balance exists)
 - `CalculateInterest()` – calculates and returns interest using formula:
$$\text{Interest} = (\text{Amount} * \text{ROI}) / 100$$
- Create multiple objects and demonstrate all methods.

Answer :

```
class BankAccount:
    ROI = 10.5
    def __init__(self, Account_holder_name, Account_balance):
        self.Account_holder_name = Account_holder_name
        self.Account_balance = Account_balance

    def Display(self):
        print(
            f"Account Holder Name : {self.Account_holder_name}\n"
            f"Current Balance : {self.Account_balance}"
        )

    def Deposit(self):
        DepositAmount = int(input("Enter Deposit Amount : "))
        self.Account_balance += DepositAmount
        print("After Deposit letest Balance is : ", self.Account_balance)

    def Withdraw(self):
        self.WithdrawAmount = int(input("Enter the Withdraw Amount : "))
        if self.WithdrawAmount <= self.Account_balance:
            self.Account_balance = self.Account_balance -
self.WithdrawAmount
            print("After Withdraw Account balance is : ", self.Account_balance)
```

```

        else:
            print("Insufficient Balance in Your Account")

    def CalculateInterest(self):
        self.Interest = (self.Account_balance * self.ROI)/100
        print("Your Saving Account Intrest is : ",self.Interest)
        self.NetBalance = self.Account_balance + self.Interest
        print("Net Balance : ",self.NetBalance)

def main():
    Obj1 = BankAccount("Amar Bhandare",100000)

    print("Please Select Operations :")
    print("1. Deposit")
    print("2. Withdraw")
    print("3. Calculate Interest")
    print("4. Display")
    print()
    Choice = int(input("Enter your choice : "))

    if Choice == 1:
        Obj1.Deposit()

    elif Choice == 2:
        Obj1.Withdraw()

    elif Choice == 3:
        Obj1.CalculateInterest()

    elif Choice == 4:
        Obj1.Display()

    else:
        print("Invalid Choice")

if __name__ == "__main__":
    main()

```

3: Write a Python program to implement a class named Numbers with the following specifications:

- The class should contain **one instance variable**:
 - Value
- Define a constructor (`__init__`) that accepts a number from the user and initializes Value.
- Implement the following **instance methods**:
 - ChkPrime() – returns True if the number is prime, otherwise returns False
 - ChkPerfect() – returns True if the number is perfect, otherwise returns False

- Factors() – displays all factors of the number
- SumFactors() – returns the sum of all factors
- Create multiple objects and call all methods.

Answer :

```
class Numbers():
    def __init__(self, Value):
        self.Value = Value

    def ChkPrime(self):
        if self.Value < 2:
            print('Not Prime')
            return False
        else:
            for i in range(2, int(self.Value ** 0.5) + 1):
                if self.Value % i == 0:
                    print("Not Prime")
                    return False
            else:
                print(f"{self.Value} is Prime Value")
                return True

    def ChkPerfect(self):
        FactAdd = 0
        for i in range(1, self.Value, 1):
            if self.Value % i == 0:
                FactAdd = FactAdd + i
        if FactAdd == self.Value:
            print(f"{self.Value} is Perfect Value")
        else:
            print("Not Perfect")
        return FactAdd == self.Value

    def Factors(self):
        Factorslist = []
        for i in range(1, self.Value + 1):
            if self.Value % i == 0:
                Factorslist.append(i)
        print("Factor are : ", Factorslist)

    def SumFactors(self):
        Factorslist = []
        for i in range(1, self.Value + 1):
            if self.Value % i == 0:
                Factorslist.append(i)
        print("SumFactor is : ", sum(Factorslist))
```

```
        return sum(Factorslist)
```

```
def main():  
    Value = int(input("Enter the Value : "))  
    Obj1 = Numbers(Value)  
    Obj1.ChkPrime()  
    Obj1.ChkPerfect()  
    Obj1.Factors()  
    Obj1.SumFactors()  
  
    Value = int(input("Enter the Value : "))  
    Obj2 = Numbers(Value)  
    Obj2.ChkPrime()  
    Obj2.ChkPerfect()  
    Obj2.Factors()  
    Obj2.SumFactors()  
  
if __name__ == "__main__":  
    main()
```